

The Class Notion in Telos, UML and Java

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Classes in knowledge representation (incl. Telos, OWL, ...)

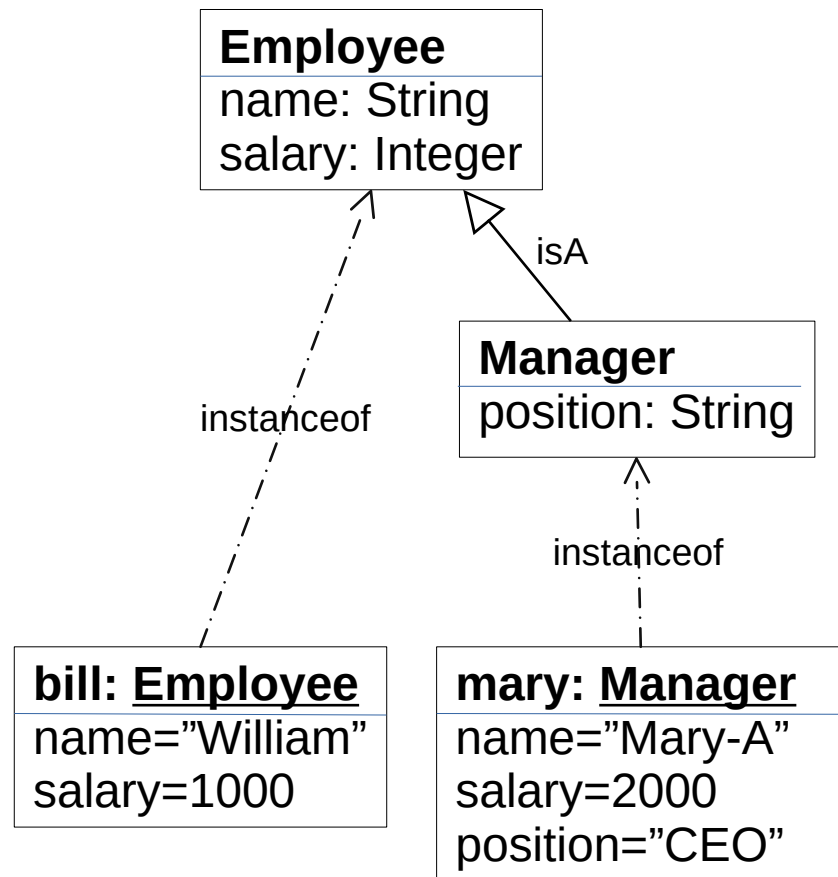
A class is a set of “individuals” (=“objects”). The class can be described extensionally by listing its elements, or intensionally by a membership condition (“all residents born between 1992 and 2003”). Type is often used as a synonym for class. For me, type is analogous to a data type. There are generally an infinite number of values of a type, but a class usually only has a finite number of objects at a given point in time.

Classes in object-oriented programming languages (incl. Java):

A class is a program module that describes the type of its "objects". The class can have class- and instance-specific variables (attributes). These have values or are references to other objects. The class defines class- and instance-specific operations. The instance-specific operations are applied to an object of the class. Each object has a state that is formed by the current values of its variables. A class can only have a finite number of objects at any one time. In Java, the class itself is also an object of the class "Class". Inheritance from superclasses is supported, but usually a class can only have one superclass. A subclass can add new attributes and methods and, depending on the programming language, also override methods of the superclasses. Which method is executed for a given object depends on the object and is decided at runtime (Java).

A data type is assigned to the variables of a class or the arguments of the methods rather than to the class itself.

Example



On the one hand, a class is an object (defined by its attributes). On the other hand, it represents its current extension (= finite set of instances).

The type of the class is the infinite set of all possible instances, i.e. for **Employee**, the set of all triples (e, n, s) from $ID \times String \times Int$.

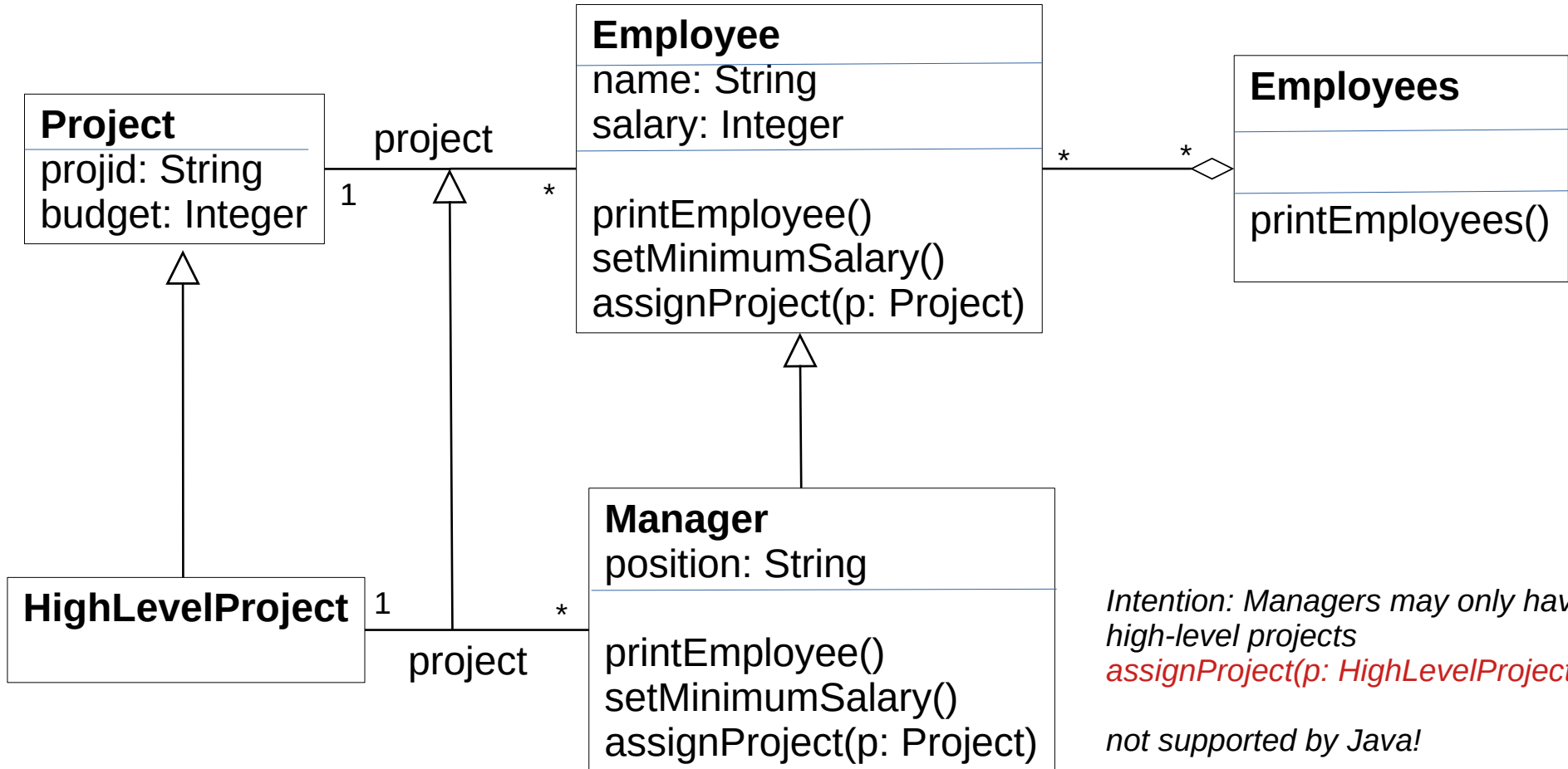
The attribute definitions of a class for objects are embodied in attribute values. The combination of attribute values is also called the state of an object.

Instances must fulfill the type condition of their class, i.e. the attribute values must be elements of the attribute types.

Questions:

- (1) Co-variant or contra-variant specialization?
- (2) Are metaclasses (classes of classes) allowed?
- (3) Multiple classification or generalization?

UML Class Diagram vs. Java



Realization with Telos/ConceptBase

